

A candidate counterexample to Conjecture 1.2 in Asymptotic structure. II. Path-width and additive quasi-isometry

github.com/graph-theory-AI — machine-generated candidate writeup

17 September 2026

Abstract

A finite counterexample is obtained from a contraction- and subdivision-closed class whose branch vertices are pairwise joined by internally degree-two paths. This candidate disproof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Conjecture 1.2](#) from *Asymptotic structure. II. Path-width and additive quasi-isometry*.

1.1 Problem statement

Conjecture. Let $\mathcal{C}$ be a class of connected graphs, closed under contracting edges and subdividing edges. For all L, C there exists C' such that if there is an (L, C) -quasi-isometry from a graph G to a member of $\mathcal{C}$, then there is a $(1, C')$ -quasi-isometry from G to a member of $\mathcal{C}$.

1.2 Catalog context

Motivated by a theorem of Chepoi, Dragan, Newman, Rabinovich, and Vaxès showing that (L, C) -quasi-isometry to a tree implies $(1, C')$ -quasi-isometry to a tree, the authors ask for which graph classes $\mathcal{C}$ such a multiplicative-to-additive distortion improvement holds. Closure under edge subdivision and edge contraction is identified as a necessary condition, and the conjecture proposes it is also sufficient. The paper proves a special case: when $\mathcal{C}$ is the class of graphs of bounded path-width.

1.3 Claimed result

Model verdict: disproved (confidence: high).

A finite counterexample is obtained from a contraction- and subdivision-closed class whose branch vertices are pairwise joined by internally degree-two paths.

The model's own caveat: Uses the standard coarsely onto definition of quasi-isometry; the class is not required to be closed under edge deletion.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 A finite counterexample

All graphs below are finite, connected, and simple. Edge contraction includes the usual suppression of loops and parallel edges.

An (L, C) -quasi-isometry $f : G \rightarrow H$ satisfies

$$L^{-1}d_G(x, x') - C \leq d_H(f(x), f(x')) \leq Ld_G(x, x') + C$$

and every vertex of H is within distance C of $f(V(G))$.

We construct a fixed class $\mathcal{C}$, closed under edge contraction and subdivision, and graphs G_g, H_g^+ such that:

- $H_g^+ \in \mathcal{C}$;
- there is an onto $(2, 1)$ -quasi-isometry $G_g \rightarrow H_g^+$;
- every $(1, A)$ -quasi-isometry from G_g to any member of $\mathcal{C}$ satisfies

$$A + 1 > \frac{g}{10000}.$$

Thus the proposed uniform additive constant does not exist.

The red/blue turn-cost mechanism from the supplied attempt is useful, but its arbitrary-map/arbitrary-target gap must be closed. We do this by adding long complete-graph corridors and proving a rigidity lemma. The infinite-graph argument is not used.

2.1.1 1. The contraction- and subdivision-closed class

For a graph J , put

$$B(J) = \{v \in V(J) : \deg_J(v) \geq 3\}.$$

A **thread** between distinct vertices $u, v \in B(J)$ is a u - v path all of whose internal vertices have degree two in J .

Define

$$\mathcal{C} = \{J : \text{every two distinct vertices of } B(J) \text{ are joined by a thread}\}. \quad (1.1)$$

In particular, every subdivision of a complete graph belongs to $\mathcal{C}$.

2.1.1.1 Lemma 1.1 The class $\mathcal{C}$ is closed under edge subdivision and edge contraction.

2.1.1.1.1 Proof Subdivision is immediate: the old branch vertices are unchanged, and threads remain threads.

For contraction, let J' be obtained from $J \in \mathcal{C}$ by contracting xy . Every branch vertex of J' is the image of a branch vertex of J . Indeed, degrees of uncontracted vertices cannot increase, and contracting two vertices of degree at most two produces a vertex of degree at most two.

Take distinct $u', v' \in B(J')$, and choose branch vertices $u, v \in B(J)$ mapping to them. Let P be a thread between u and v in J .

The image of P contains a u' - v' path whose internal vertices have degree at most two in J' . To see the only potentially problematic point, suppose the contracted vertex occurs internally on this image. If one endpoint of xy is internal on P , its two neighbours already lie on P . Hence the other endpoint

of xy also lies on P ; unless the contracted vertex is an endpoint of the image, both contracted vertices have degree two. Their contraction therefore has degree at most two.

After choosing a simple path in the image, its internal vertices have degree exactly two. Thus u', v' are joined by a thread. $\square$

2.1.2 2. Metric preliminaries

We use the geometric realization of a graph, giving every edge length one. Distances between original vertices are unchanged.

2.1.2.1 Lemma 2.1 — Continuous coarse inverses If there is a $(1, A)$ -quasi-isometry between graphs G and K , then their geometric realizations X, Y admit continuous maps

$$F : X \rightarrow Y, \quad Q : Y \rightarrow X$$

such that, with

$$D = 10(A + 1), \quad (2.1)$$

we have

$$\begin{aligned} |d_Y(Fx, Fx') - d_X(x, x')| &\leq D, \\ |d_X(Qy, Qy') - d_Y(y, y')| &\leq D, \\ d_Y(FQy, y) &\leq D. \end{aligned} \quad (2.2)$$

2.1.2.1.1 Proof Extend the given vertex map over each edge along a geodesic. The resulting continuous map F has distortion at most $3A + 4$: each point can be compared with an endpoint of its edge, and each image edge has length at most $A + 1$.

For every vertex w of K , choose a vertex $q(w)$ of G with

$$d_K(f(q(w)), w) \leq A.$$

The vertex map q has distortion at most $3A$. Extend it over edges along geodesics. The resulting Q has distortion at most $9A + 4$.

Finally, comparing y with an endpoint w of its edge gives

$$d_Y(FQy, y) \leq 7A + 6.$$

All three bounds are at most $10(A + 1)$. $\square$

We also use the following elementary observation.

2.1.2.2 Lemma 2.2 — Detecting a branch vertex Let Z be a metric graph. Suppose z_1, z_2, z_3 lie within distance r of z , and

$$d_Z(z_i, z_j) > r \quad (i \neq j).$$

Then $B(Z)$ meets the closed radius- r ball about z .

2.1.2.2.1 Proof Otherwise, the union of shortest z - z_i paths has no branch vertex. It lies in a path, or forms a cycle of length at most $2r$. In the path case, two of the three points lie on the same side of z , at distance at most r from each other. In the cycle case, the diameter is at most r . Both contradict the hypothesis. $\square$

2.1.3 3. Rigidity of a complete system of long corridors

Here is the step that rules out changing either the quasi-isometry or the target.

2.1.3.1 Lemma 3.1 — Long-corridor rigidity Let $n \geq 4$, $D \geq 1$, and $S \geq 1000D$.

Construct a metric graph X as follows:

- replace each vertex v of K_n by a tree T_v of diameter at most one;
- choose $r_v \in T_v$;
- for each pair u, v , join a point of T_u to a point of T_v by one corridor I_{uv} , of length

$$L_{uv} \geq S;$$

- these corridors have disjoint interiors, and there are no other edges.

Let Y be the geometric realization of a graph in $\mathcal{C}$. Suppose continuous maps $F : X \rightarrow Y$, $Q : Y \rightarrow X$ satisfy (2.2).

Then:

1. Y has exactly n branch vertices, labelled h_v ;
2. for each pair u, v , there is exactly one thread between h_u, h_v ;
3. writing λ_{uv} for its length, the weighted complete-graph metric d_λ satisfies

$$|d_\lambda(u, v) - d_X(r_u, r_v)| \leq 30D; \quad (3.1)$$

4. every such thread satisfies

$$\lambda_{uv} \geq \frac{L_{uv}}{2} - D. \quad (3.2)$$

2.1.3.1.1 Proof

2.1.3.2 Step 1: Find a branch vertex near each $F(r_v)$ Choose three different corridors incident with T_v , and on each choose a point at corridor-distance $10D$ from T_v . Each point is within $10D + 1$ of r_v , and each pair is at distance at least $20D$.

Their images under F lie within $12D$ of $F(r_v)$, and have pairwise distances at least $19D$. Lemma 2.2 therefore gives

$$h_v \in B(Y), \quad d_Y(h_v, F(r_v)) \leq 12D. \quad (3.3)$$

For distinct u, v , we have $d_X(r_u, r_v) \geq S$, so

$$d_Y(h_u, h_v) \geq S - 25D. \quad (3.4)$$

In particular, these n branch vertices are distinct.

2.1.3.3 Step 2: Every branch vertex of Y belongs to one of these clusters Let $h \in B(Y)$. By (3.4), at most one of the chosen h_v lies within distance $S/3$ of h . Since $n \geq 4$, choose three others, each at distance at least $S/3$.

The defining property of $\mathcal{C}$ supplies threads from h to these three vertices. The threads have disjoint interiors, and each has length at least $S/3$. Points at thread-distance $10D$ from h therefore have distance $10D$ from h and pairwise distance $20D$.

Their images under Q , together with Lemma 2.2, show that $Q(h)$ lies within $11D$ of a branch vertex of X . All branch vertices of X lie in the trees T_v . Consequently,

$$d_X(Q(h), r_v) \leq 12D \quad (3.5)$$

for some v . Using (2.2),

$$d_Y(h, F(r_v)) \leq 14D. \quad (3.6)$$

This v is unique because distinct $F(r_v)$ are at distance at least $S - D > 28D$.

Call it the cluster of h . The initially chosen h_v belongs to cluster v .

2.1.3.4 Step 3: A thread between two clusters must follow the corresponding corridor

Consider any thread P between branch vertices h, h' in distinct clusters v, w . Its length λ satisfies

$$\lambda \geq d_Y(h, h') \geq S - 29D > 100D.$$

Remove the first and last $50D$ of P , leaving a closed interval P^0 .

Every point $y \in P^0$ has distance at least $50D$ from $B(Y)$, since the interior of a thread can be left only through its endpoints.

If $d_X(Qy, r_z) \leq 20D$, then (2.2) and (3.3) give

$$d_Y(y, h_z) \leq D + (20D + D) + 12D = 34D,$$

a contradiction. Thus

$$Q(P^0) \cap \bigcup_z \bar{B}_X(r_z, 20D) = \emptyset. \quad (3.7)$$

The components of the complement in (3.7) are precisely trimmed interiors of the individual corridors of X . Since Q is continuous, $Q(P^0)$ lies in one such component.

Let y_0, y_1 be the two endpoints of P^0 . By (3.5),

$$d_X(Qy_0, r_v) \leq 50D + D + 12D = 63D,$$

and similarly $d_X(Qy_1, r_w) \leq 63D$. Since $63D < S$, the corridor containing $Q(P^0)$ must be incident with both v and w . It is therefore I_{vw} .

Moreover, measured from the respective ends of I_{vw} , the points Qy_0, Qy_1 lie within $63D$. By continuity, $Q(P^0)$ covers the whole central interval

$$[63D, L_{vw} - 63D] \quad (3.8)$$

of that corridor.

2.1.3.5 Step 4: No two threads can join the same two clusters If two distinct threads joined clusters v, w , (3.8) would give interior points y, y' , one on each trimmed thread, mapping under Q to the midpoint of I_{vw} .

The distortion bound for Q would imply

$$d_Y(y, y') \leq D.$$

But distinct threads have disjoint interiors, and leaving the first trimmed thread requires travelling at least $50D$. This is impossible.

Every cluster is nonempty. If some cluster contained two branch vertices, threads from both of them to a branch vertex in any other cluster would contradict the preceding paragraph. Thus every cluster contains exactly one branch vertex.

The defining property of $\mathcal{C}$ now gives exactly one thread between each pair h_u, h_v .

2.1.3.6 Step 5: The metric and thread-length conclusions Any shortest path between branch vertices traverses a sequence of the threads just identified. Other components—pendant paths or cycles attached at a single branch vertex—cannot shorten such a path. Therefore their metric is exactly the weighted K_n metric with edge lengths λ_{uv} .

Equation (3.3) and the distortion of F give

$$|d_Y(h_u, h_v) - d_X(r_u, r_v)| \leq 25D,$$

which implies (3.1).

Finally, (3.8) includes the quarter and three-quarter points of I_{uv} . Their distance in X is $L_{uv}/2$: the direct interval has that length, and a route through the two ends cannot be shorter. Choosing preimages on the thread and using the distortion of Q yields

$$\lambda_{uv} \geq L_{uv}/2 - D.$$

This proves (3.2). $\square$

2.1.4 4. High-girth graphs with two colours

For every integer $g \geq 5$, we need a finite connected simple graph H_g of girth at least g , with

$$E(H_g) = E_R \sqcup E_B,$$

such that every vertex has two incident red edges and two incident blue edges.

Here is a self-contained construction.

Let W_g be the finite set of all nonempty freely reduced words of length less than g in

$$a, a^{-1}, b, b^{-1}.$$

For a word w of length k , take the set $\{0, \dots, k\}$, and prescribe partial permutations a_w, b_w so that following the letters of w sends

$$0 \rightarrow 1 \rightarrow \dots \rightarrow k.$$

For an inverse letter, prescribe the corresponding reversed permutation arrow. These partial permutations are well-defined and injective: a conflict at an intermediate index would require adjacent inverse letters. Complete each partial permutation to a permutation, for example by matching the remaining domain and range elements in increasing order.

In the product of these finite symmetric groups, let a, b be the tuples of the constructed permutations, and let Q_g be the subgroup they generate. No nonempty freely reduced word of length less than g is trivial in Q_g , because its own coordinate sends 0 to k .

The undirected Cayley graph of Q_g with generators $a^{\pm 1}, b^{\pm 1}$ is therefore simple, connected, 4-regular, and has girth at least g . Colour a -edges red and b -edges blue.

Each monochromatic subgraph is a union of cycles, all of length at least g .

2.1.5 5. The graphs and their uniform quasi-isometries

Fix an integer $g \geq 10000$, and write

$$H = H_g, \quad n = |V(H)|, \quad S = g, \quad M = 100g^2. \quad (5.1)$$

Construct H_g^+ from K_n , on vertex set $V(H)$, by replacing:

- every edge of H by a path of length S ;
- every edge not in H by a path of length M .

Thus H_g^+ is a subdivision of K_n , so

$$H_g^+ \in \mathcal{C}.$$

Construct G_g by splitting each original vertex v into two vertices v_R, v_B , joined by a unit-length **switch edge**. Attach:

- the red H -corridors to the corresponding R -vertices;
- the blue H -corridors to the corresponding B -vertices;
- every length- M corridor to the corresponding R -vertices at both ends.

Define

$$p : G_g \rightarrow H_g^+$$

by collapsing each switch edge and mapping all corridor vertices to their corresponding vertices.

This map is onto. Every G_g -path projects to an H_g^+ -walk of no greater length, so

$$d_{H_g^+}(p(x), p(y)) \leq d_{G_g}(x, y).$$

Conversely, a shortest H_g^+ -path of length k lifts using at most $k + 1$ switches. Hence

$$d_{G_g}(x, y) \leq 2d_{H_g^+}(p(x), p(y)) + 1.$$

Therefore p is an onto $(2, 1)$ -quasi-isometry.

2.1.6 6. No uniformly additive approximation exists

Suppose there is a $(1, A)$ -quasi-isometry

$$G_g \rightarrow K, \quad K \in \mathcal{C},$$

where

$$A + 1 \leq \frac{g}{10000}. \quad (6.1)$$

Put

$$D = 10(A + 1), \quad E = 30D, \quad s = \left\lfloor \frac{g}{10} \right\rfloor. \quad (6.2)$$

Then $S = g \geq 1000D$, so Lemmas 2.1 and 3.1 apply to G_g , with the switch edges as the trees T_v and $r_v = v_R$.

We obtain positive lengths λ_{uv} on the edges of K_n such that

$$|d_\lambda(u, v) - d_{G_g}(u_R, v_R)| \leq E \quad (6.3)$$

for every u, v , and

$$\lambda_{uv} \geq M/2 - D \quad \text{if } uv \notin E(H). \quad (6.4)$$

The following inequalities follow from $g \geq 10000$ and $D \leq g/1000$:

$$\begin{aligned} s - 3 &> 2E, & Ss &> 2 + 2E, & S &> s + 1, \\ M/2 - D &> Ss + 2 + E, & 2s &< g/2, & g - s &\geq 2s. \end{aligned} \quad (6.5)$$

We now show that (6.3)–(6.4) are impossible.

2.1.6.1 6.1. Lower bounds on short H -paths If P is an H -path of length $t < g/2$, then P is geodesic in H . Let its endpoints be u, v .

Any G_g -path from u_R to v_R that uses a long corridor has length at least $M > St$. A path using only H -corridors has length at least St . Thus

$$d_{G_g}(u_R, v_R) \geq St.$$

Consequently,

$$\lambda(P) \geq d_\lambda(u, v) \geq St - E. \quad (6.6)$$

2.1.6.2 6.2. Upper bounds on monochromatic paths Let P be a monochromatic H -path of length s , with endpoints u, v .

If P is red, it lifts from u_R to v_R with length Ss . If it is blue, two endpoint switches suffice. Hence

$$d_{G_g}(u_R, v_R) \leq Ss + 2,$$

and so

$$d_\lambda(u, v) \leq Ss + 2 + E. \quad (6.7)$$

By (6.4)–(6.5), a λ -shortest path Q between u, v cannot use an edge outside H . Choose Q simple.

If $Q \neq P$, their union contains a cycle, giving

$$|Q| + s \geq g.$$

In particular, Q contains a $2s$ -edge subpath. Applying (6.6) to that subpath gives

$$\lambda(Q) \geq 2Ss - E > Ss + 2 + E,$$

contrary to (6.7). Thus P itself is shortest, and

$$\lambda(P) \leq Ss + 2 + E. \quad (6.8)$$

2.1.6.3 6.3. Average over the monochromatic cycles Apply (6.8) to every cyclic window of s edges on every monochromatic cycle. Each edge of a cycle occurs in exactly s windows. Summing over both colours gives

$$\sum_{e \in E(H)} \lambda_e \leq |E(H)| \left(S + \frac{2 + E}{s} \right). \quad (6.9)$$

Now choose a uniformly random directed edge of H , and continue for s edges by alternating colours, choosing uniformly between the two available edges of the required colour.

At every position the directed edge is uniformly distributed: every directed edge has exactly two possible predecessor directed edges of the opposite colour, each transitioning to it with probability $1/2$.

Thus (6.9) implies that the expected λ -length of this walk is at most

$$Ss + 2 + E.$$

There is therefore an alternating walk

$$P = v_0 v_1 \cdots v_s$$

with

$$\lambda(P) \leq Ss + 2 + E. \quad (6.10)$$

It is nonbacktracking. Since $s < g$, it is a path; since $2s < g$, it is the unique H -geodesic between its endpoints.

2.1.6.4 6.4. Alternating paths require accumulated switch costs Following P in G_g , from $(v_0)_R$ to $(v_s)_R$, costs at most

$$Ss + s + 1.$$

A shortest path cannot use a long corridor, since M exceeds this quantity.

Nor can it use at least $s + 1$ short corridors, since

$$S(s + 1) > Ss + s + 1.$$

It must therefore use exactly s short corridors and project to the unique H -geodesic P .

Because P alternates colours, a switch is necessary between every two successive corridors. Consequently,

$$d_{G_g}((v_0)_R, (v_s)_R) \geq Ss + s - 1. \quad (6.11)$$

Combining (6.3), (6.10), and (6.11),

$$Ss + s - 1 - E \leq d_\lambda(v_0, v_s) \leq \lambda(P) \leq Ss + 2 + E.$$

Therefore

$$s - 3 \leq 2E,$$

contradicting (6.5).

This proves that (6.1) is impossible.

2.1.7 7. Conclusion and scope

For the fixed class $\mathcal{C}$ defined in (1.1), every $g \geq 10000$ gives an onto $(2, 1)$ -quasi-isometry

$$G_g \rightarrow H_g^+ \in \mathcal{C},$$

but every $(1, A)$ -quasi-isometry from G_g to any member of $\mathcal{C}$ satisfies

$$A + 1 > \frac{g}{10000}.$$

Hence the conjecture, with exactly the closure assumptions in the question, is **false for finite graphs**.

The crucial addition to the supplied attempt is Lemma 3.1: it turns an arbitrary additive quasi-isometry to an arbitrary member of $\mathcal{C}$ into edge lengths on the prescribed complete-graph skeleton, with only a constant-factor increase in additive error. The long corridors enforce this rigidity; the red/blue averaging then supplies the contradiction.

No infinite-graph convention or external conjecture is needed. A version imposing the additional requirement of closure under edge deletion is not addressed by this counterexample.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	2509.09031__00
Catalog entry	Conjecture 1.2
Source corpus	arxiv
Campaign leg	selected second attempts (attacks_retry)
Source paper / entry	Asymptotic structure. II. Path-width and additive quasi-isometry
Model verdict	disproved (confidence: high)
Selection criterion	would_publish: true and verdict proved or disproved
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_retry/2509.09031__00/output.md
Independent review	None
arXiv	arXiv:2509.09031
Previous attempt	attacks/2509.09031__00: gpt-5.6-sol verdict partial
Model's caveats	Uses the standard coarsely onto definition of quasi-isometry; the class is not required to be closed under edge deletion.

The generated Markdown and LaTeX sources for this PDF are stored under `to_review_astra/src/2509.09031__00/`.